

REPORT

Authentication system And Managements

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I. INTRODUCTION

1. Problem Statement

As we can see, many people around the world meet different types of social engineering attacks. Some people have limited knowledge in this technology field and don't know how to mitigate this problem. Most people are getting their social media accounts hacked by attackers due to weak passwords, no authentication, no encryption platform, and weak security platforms.

2. Objective

This project aims to enhance the security login system in order to mitigate modern issues and provide trust to all users by using new technology to prevent hackers from sneaking into the system. Moreover, this project hides user credential information to maintain privacy for all users, ensuring that even the owner and developers do not know the user information. The project aims to be implemented as a real-world solution to alleviate the risk of attacks. As we all know, prevention is better than cure.

3. Brief Project Description

This project will enhance the security system by implementing each functionality precisely in order to provide trust to all users. Each functionality in this project has a different security system and is user-friendly as well. To mitigate modern issues, we have been using some new technology, such as authentication, encryption, and password masking, which enhance the security system to store each user's credential information without concern. This project doesn't allow any user to spam username and password by blocking the spammer and letting them exit the system permanently.

II. METHODOLOGY

1. Tools, Technology and Libraries used

❖ Tools



Is a web-based platform that facilitates real-time collaboration, allowing teams to work together on code development, building web pages, and updating project content. It is particularly useful for version control, tracking changes, and managing contributions in both small and large projects.



Is a video conferencing service developed by google. it allows you to use for video and audio to connect with others over the internet, used for team discussions and project work or meeting.



Visual Studio Code

Is a streamlined code editor with support for development operations like debugging, task running, and version control. Also, vs code can be used with a variety of programming languages, including C, C#, C++, Fortran, Go, Java, JavaScript, Node.js, Python, Rust, and Julia.



is a secure messaging and audio calling cloud-based app that allows you to send, messages, photos, videos, voice and any files of type to your contacts for free. It is available for Android, iOS, Windows, macOS, Linux, and web browsers. Telegram offers end-to-end encryption in voice and video calls, and in optional private chats, which Telegram calls Secret Chats.



is a word processing program developed by Microsoft. Microsoft Word is a text processing application used to create, edit, and format text documents. It offers features like templates, spell check, and collaboration tools, making it ideal for creating project reports, proposals, and other professional documents.



Canva is an online design and visual communication platform with a mission to empower everyone in the world to design anything and publish anywhere. Canva helps you design everything from posts, flyers, and stories to resumes, magazine covers, banners, logos, and presentations instantly.

❖ Technology



Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built-in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development. Python's simple, easy-to-learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages for modularity and code reuse. The Python interpreter and extensive standard library are freely available for all major platforms.

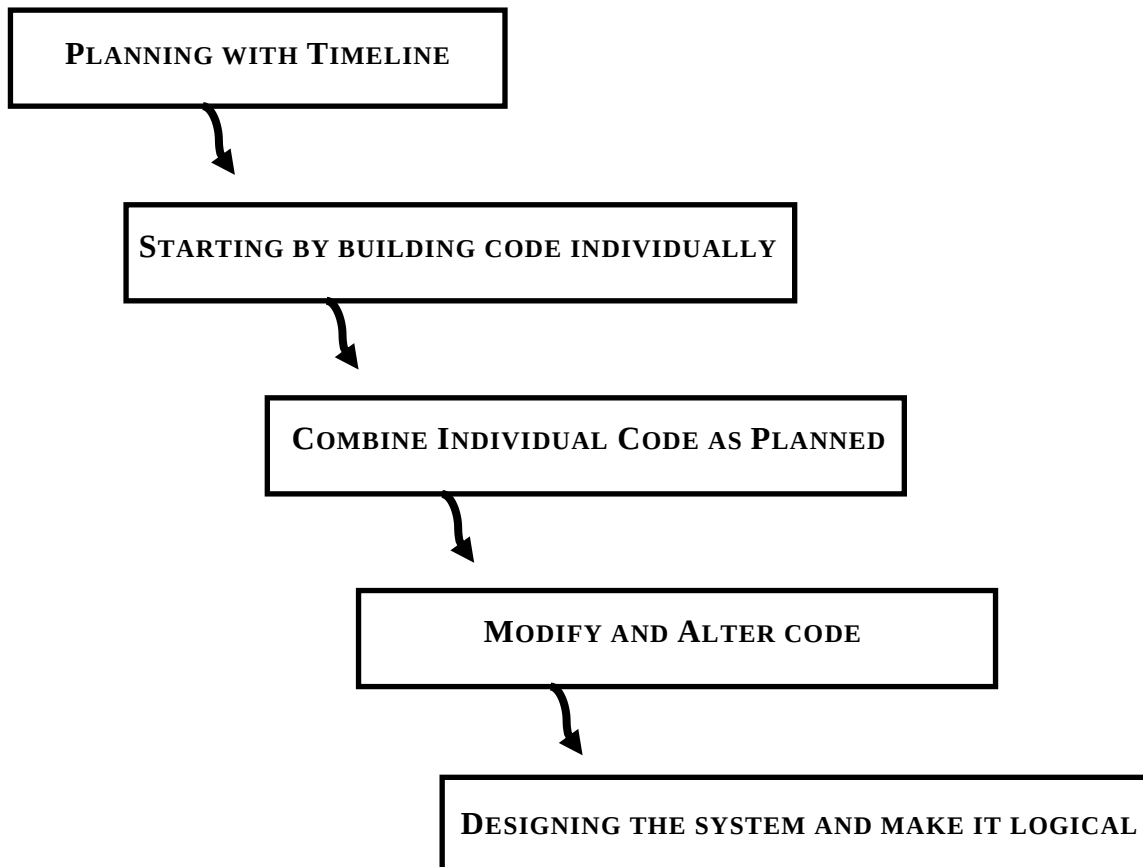
❖ Libraries

- **os (Operation System)** library in Python allows use to interact (see, move, create, and delete) with the operating system.
- **json (JavaScript Object Notation)** library in Python use to store and share the data.
- **sys (system)** library is a tool in Python that provides access to variables and functions that interact with the Python interpreter.
- **cryptography** library in Python is a tool that helps us protect and secure data by making it unreadable to anyone who doesn't have the right key.
- **random** library in Python is a tool that helps us work with random numbers and make random choices.
- **maskpass** library in Python is a tool that helps us securely hide user input when they type something, like a password.
- **time** library is a useful tool in Python that helps you work with time. You can use it to get the current time, pause your program for a certain amount of time (like waiting for a few seconds), measure how long something takes to run, format the time in a way that's easy to read.
- **re (regular expression)** library in Python helps us to find and work with text in very flexible and powerful ways by using patterns, which can make tasks like searching and editing text much easier.

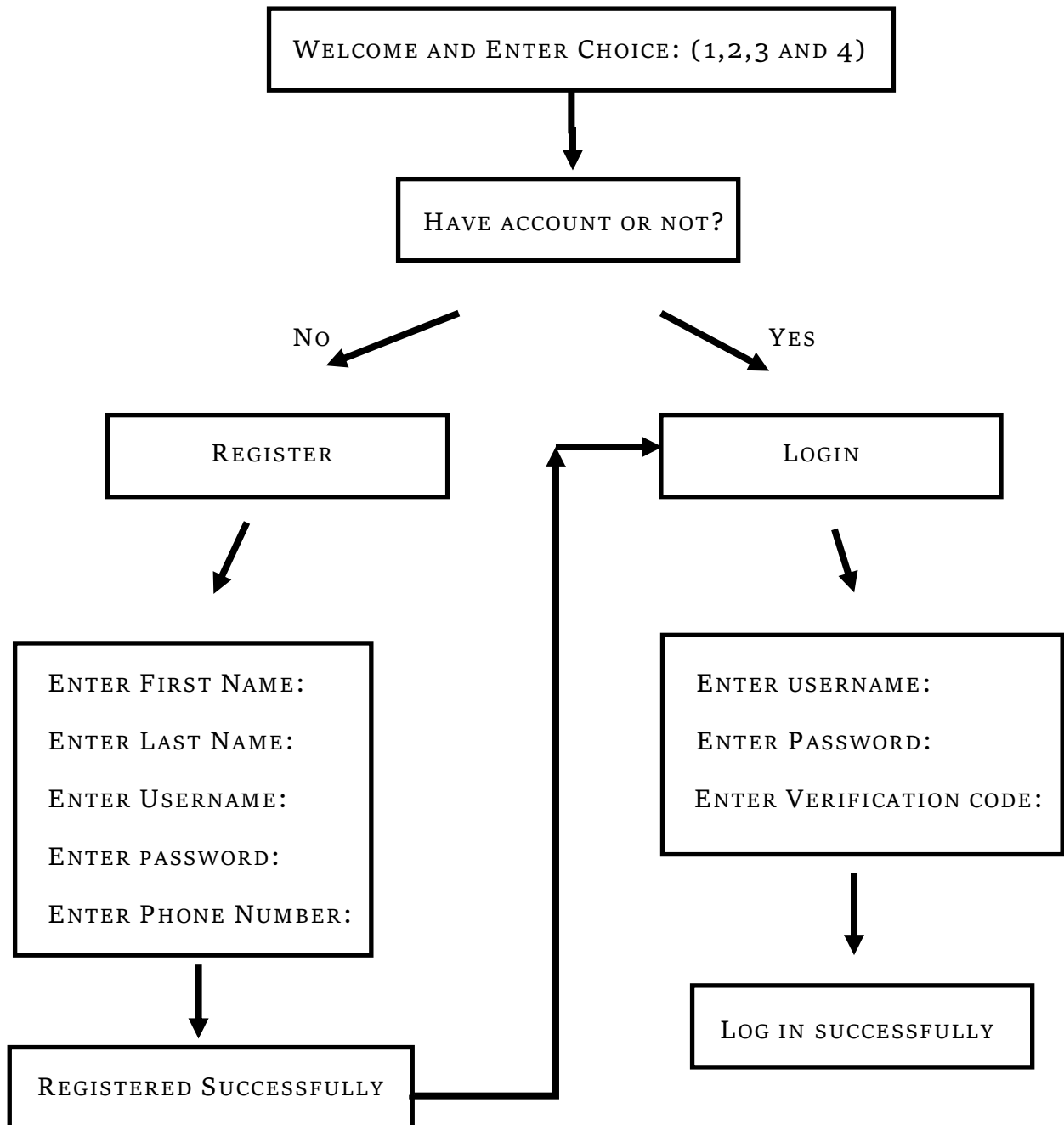
2. Process / Workflow

Use a simple flow diagram or list the key steps in the development process.

FLOW-DIAGRAM OF PROJECT DEVELOPMENT



FLOW OF CODE TO LOGIN SUCCESSFULLY



III. IMPLEMENTATION

1. Key Features

1.1. Registration

The registration system is designed to securely create a user account by providing necessary details, such as first name, last name, password, phone number. The system is designed to ensure the confidentiality, integrity, and authenticity of user data through techniques like input validation, encryption, and secure communication protocols. Below is a detailed description of its functionality.

Functionality Overview:

❖ **User account:**

- First name and last name: Captures the user's personal identification details.
- Username: allow users to create a unique username that can be used for login features.
- Password: A strong password is required for securing the account.
- Phone number: Used for additional verification and as a recovery option; it is validated to ensure correctness.

❖ **Input Validation:**

- Uniqueness of Username: Ensure that the username is not already registered.
- Phone number: verifies that the phone number follows the standard structure (e.g., +855 12123456).

➤ **Password Strength:**

- ✓ must be minimum 8 characters
- ✓ include at least one uppercase letter, one lowercase letter, one number
- ✓ Prevents the use of commonly used or weak passwords.

❖ **Data storage:**

User information is stored securely in a structured format, such as:

- JSON file: Simple key-value storage also similar to dictionary in python.
- Text File: In smaller-scale or prototype applications, user data may be stored in a simple text file (e.g., .txt).

1.2. Login

- ❖ **Invisible password:** using maskpass library to hide password when input.
- ❖ **User friendly:** using file and error handling to make sure nothing interrupt user when they using our program
- ❖ **Verify code:** using code for verification is the best way to secure user's information. We use random library to generate 6 digits code for verification step.
- ❖ **Attempt limited:** limit the system at most 3 attempts to prevent Bruce force attack.
- ❖ **Time cooldown:** Using this method can make users more careful before doing something. (using time library)

1.3. Password Recovery

- ❖ **User verification:** system asks for **username**.
- ❖ **Attempt:** Allows users to try up to **3 times** to provide correct information before ending the process.
- ❖ **Password Reset:** Users can **reset their password** after enter their username correctly.
- ❖ **Error Handling:** The system gives **clear messages** if you enter something wrong or if your passwords don't match.
- ❖ **User options:** If username not found in database, user can:
 - **Register:** If they want to create a new account.
 - **Login:** If they remember their password.
 - **Password Recovery:** If they remember their username.
 - **Exit:** If they want to leave.
- ❖ **Confirmation Steps:** Users must **confirm the new password** to avoid mistakes.

1.4. Encryption

using fernet under the umbrella of cryptography, which create a key to use it for encrypt and decrypt function while this key was generated and stored it in another file name "pw.key". This encryption is applied into the project such as Register, Login, Password Recovery.

- ❖ For register, we use encryption to encrypt the password when it is sent to be stored in file, and making the password unreadable that make the user credential information safe.
- ❖ For Login, we use decryption to decrypt the password and check if the password input matching with the decrypted password.
- ❖ For Password Recovery, we apply the encryption method as in register after users change their new password and this password will be encrypted to store in the database.

IV. RESULT AND TESTING

1. Output

○ Main Page

```
+=====+
|                                     |
|          * * * * *                 |
|        A U T H E N T I C A T I O N |
|          S Y S T E M               |
|          &                        |
|        M A N A G E M E N T S       |
|                                     |
+=====+

Please choose one of the following options:

1. Register
2. Login
3. Password Recovery
4. Exit

Enter your choice:
```

○ Registration Page

```
Enter your choice: 1

=====
|           Welcome To REGISTRATION SYSTEM           |
=====

Please follow the instructions to register a new account.

First name: Aphinun
Last name: Romnea
Username: APN
Username accepted!
Enter your password: Aphinun123@
Password strength: Strong
Confirm your password: Aphinun123@
Password set successfully!
Phone number: +855 78283848
User data saved to 'database.txt'
Registered Successfully!

Registration Complete!
Thank you for joining our system!

=====
```

- **Login Page**

```
Enter your choice: 2

=====
|           Welcome To LOGIN SYSTEM           |
=====
Welcome to the Login System!
Enter username: APN
Enter Password:
Your verification code is:2WPXNZ
Enter code: 2WPXNZ
You have successfully logged in to your account!
=====
```

- **Password Recovery Page**

```
Enter your choice: 3

=====
| Welcome To PASSWORD RECOVERY SYSTEM |
=====
Enter Username: APN
User found! You can now reset your password.
Enter a new password: Aphinun123!
Confirm your new password: Aphinun123!
Password updated successfully!
```

2. Testing Methods

- ❖ **Scenario 1**

- Testing by using same username and it will exit the program definitely.

```
=====
|           Welcome To REGISTRATION SYSTEM           |
=====

Please follow the instructions to register a new account.

First name: Sophat
Last name: Vann
Username: APN
Username already taken, please choose a different one.
Attempts remaining: 2
Username: APN
Username already taken, please choose a different one.
Attempts remaining: 1
Username: APN
Username already taken, please choose a different one.
Attempts remaining: 0
Too many Attempts. Registration Failed.
```

- Testing the password, which type of password allow to register

```
=====
|           Welcome To REGISTRATION SYSTEM           |
=====

Please follow the instructions to register a new account.

First name: Sophat
Last name: Vann
Username: SV
Username accepted!
Enter your password: sophat
Password strength: Weak
Error: Password is too weak. Use a stronger password.
Enter your password: Sophat123
Password strength: Moderate
Error: Password is Moderate. Use a stronger password.
Enter your password: Sophat123@
Password strength: Strong
Confirm your password: Sophat123@
Password set successfully!
Phone number: +855 16273748
User data saved to 'database.txt'
Registered Successfully!

Registration Complete!
Thank you for joining our system!

=====
```

❖ Scenario 2

- Testing by entered wrong password more than 3 attempts; the system blocks the user and redirect to main page.

```
=====
|           Welcome To LOGIN SYSTEM           |
=====

Welcome to the Login System!
Enter username: APN
Enter Password:
!!! Wrong password !!!
!!! Please try again !!!
Remaining attempts: 2
Enter Password:
!!! Wrong password !!!
!!! Please try again !!!
Remaining attempts: 1
Enter Password:
!!! Wrong password !!!
!!! Please try again !!!
Remaining attempts: 0
Please wait before trying again.
Cooldown time remaining: 5 seconds
```

❖ Scenario 3

- When recovering a password, if a wrong username that is not in the system is entered, the system allows 3 attempts before redirecting to the main page.

```
=====
| Welcome To PASSWORD RECOVERY SYSTEM |
=====
Enter Username: Aps
User not found.
Would you like to [register], [retype], or [leave]? retype
Remaining attempts: 2
Enter Username: ajs
User not found.
Would you like to [register], [retype], or [leave]? retype
Remaining attempts: 1
Enter Username: ksdj
User not found.
Would you like to [register], [retype], or [leave]? retype
Remaining attempts: 0
Too many failed attempts. Returning to the main page.
```

- When recovering a password, if the user enters a username that exists in the system but provides a password that is not strong enough and is considered spam, the system will redirect them to the main page.

```
=====
| Welcome To PASSWORD RECOVERY SYSTEM |
=====
Enter Username: APN
User found! You can now reset your password.
Enter a new password: jkdfjksdj
Password is not strong enough. Use a combination of uppercase, lowercase, numbers, and special characters.
Remaining attempts: 2
Enter a new password: njsfjds
Password is not strong enough. Use a combination of uppercase, lowercase, numbers, and special characters.
Remaining attempts: 1
Enter a new password: ndjjs
Password is not strong enough. Use a combination of uppercase, lowercase, numbers, and special characters.
Remaining attempts: 0
Too many failed attempts. Returning to the main page.
```

V. CONCLUSION

Summarize the project outcomes:

1. Lesson Learned

- ❖ **Data Structure:** We have learned and understood about dictionary, and list more specifically. Moreover, we can use them more technically by accessing items, adding, removing, updating, and adjusting it well.
- ❖ **OPP:** by building this project, we know how each component in OOP works, such as encapsulation is used for keep our password private, inheritance is used to reduce the usability of the code and flexibility, and data abstraction is used to hide all the complex code from users and other developers
- ❖ **Libraries:** We have learned a bunch of libraries in the python such as, Fernet, which is used to encrypt and decrypt text and it is under the umbrella of cryptography; OS, which is used to check location file or interacts with file, such as create, move, delete, and etc.
- ❖ **Cybersecurity Concepts:** We have learned many different things in this field, by knowing how to mitigate the problem of how attackers sneak into the server and the system. Moreover, we have been applying many strategies to alleviate the risk of attacking. Therefore, we understand deeply on what does it exists in this field.

2. Challenges Faced

- ❖ **Building Code Individually:** we have some problems on building each functionality, because we all are new to this type of project such as new library, which we never used it before and didn't understand the process of how to use properly while we have spent couple of days to figure it out.
- ❖ **Combine Individual code into ONE:** we have many problems on this term by having couple issues of file accessing, different variables, and different functions, which made it difficult to combine everything into one project. Therefore, we had to start by testing each function in each code, and ensure it can work with each other simultaneously.
- ❖ **Modify and Alter Code:** After we made it work together, we changed some functions and classes and imported functions from other files to reduce code redundancy. However, we encountered various issues, such as files not being imported between folders, module errors, and file not found errors. As a result, all members of our team struggled with these problems, but after conducting research, we discovered a library called sys that helped us solve them.

3. Future Work

- ❖ **GUI (Graphic User Interface):** we can create a user interface in order make our program more user friendly and attractive.
- ❖ **Authentication:** In the future, we will add more security system into our program in order to minimize the risks of attacking from attackers, and to ensure that our users credential information is safe with us.
- ❖ **Scalable:** If our program works smoothly and gaining users, we will extend and enhance it in order to maintain the functionality remains efficient.

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